Augmented GRU with learned feature encoders

Candidate g112 | representative seed 3 (lowest validation loss) | restored epoch 18 | 40,945 parameter entries

A Encode inputs available before response t

Log(1 + student time gap in seconds)
1 standardized numeric value

Time encoder
Linear 1 → 32; ReLU
Linear 32 → 16
16 learned time values

4 standardized LKT inputs
intercept; student-item logitdec;
student-item PPES; student logitdec

History-feature encoder
Linear 4 → 32; ReLU
Linear 32 → 16
16 learned history values

Current item ID: KC..Default.
144 known items

Learned item embedding
145 × 16 table
144 known + 1 reserved zero row
16 item values

Current skill ID: KC..Cluster.
36 known skills

Learned skill embedding
37 × 16 table
36 known + 1 reserved zero row
16 skill values

Concatenate: 16 time + 16 history + 16 item + 16 skill = 64 coordinates
Input dropout 0.1 produces x(t), shared by prediction and memory update

Each encoder has its own weights and biases. Its 16-value output is linear; ReLU is applied after the 32-unit hidden layer.

B Predict the current response

Previous top-layer memory

Previous h3(t−1): 32
+ current x(t): 64
96 coordinates

Linear 96 → 128 (with bias)
ReLU
Head dropout 0.1

Linear 128 → 1
(with bias)
One logit

Sigmoid + clipping
p(correct at trial t)
ε = 10⁻⁵

The current answer is still unknown here. The head reads the previous top-layer memory, just as in the original model.

C Observe and encode the answer, then update memory

Observed response y(t)
0 = incorrect; 1 = correct

Response encoder
Linear 1 → 32; ReLU
Linear 32 → 16
16 learned answer values

x(t): 64

Concatenate predictors + answer
64 + 16 = 80 coordinates

h1(t−1)
32 previous values

h2(t−1)
32 previous values

h3(t−1)
32 previous values

h1(t): 32

GRU layer 1
32 units
Own state

Dropout
0.1

h2(t): 32

GRU layer 2
32 units
Own state

Dropout
0.1

GRU layer 3
32 units
Own state

h3(t)
32 values

Retain all three states for the next trial: [h1(t), h2(t), h3(t)] = 3 × 32 values

Each new participant starts with zero memory. Updated memory affects the next prediction, p(t+1).

All encoders, embeddings, GRU layers and head weights are learned jointly by backpropagation; continuation keeps weights fixed.

Encoder dropout is zero. Input, between-GRU-layer and head dropout are 0.1 during training and disabled during evaluation.